

Unifying spatially-explicit Rao's entropy, spatial concentration, and accumulation curves: the different aspects of local biodiversity

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Abstract

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1. Introduction

Local biotic and abiotic interactions are for long recognized as fundamental processes that shape the structure of ecological communities. In particular, the spatial pattern of plant communities may result from the combined effects of seed dispersal, inter-individual (i.e. intra- and inter-specific) competition/facilitation and habitat preferences (Dale 1999). While each single process may either pro-

mote or demote species coexistence, the diversity of interactions in space and/or time is a key feature for the maintenance of biological diversity (Clark et al. 2010; Cadotte et al. 2012). As a consequence, the ecological literature reveals increasing efforts in integrating approaches from phylogenetics, functional and community ecology (Mouquet et al. 2012), in order to account for the key components of species interactions with their biotic and abiotic environments.

Towards this goal, a number of novel analytical methods incorporating measures of between-species phylogenetic relatedness or functional trait dissimilarities as a surrogate for ecological differences have been developed (Pavoine and Bonsall 2011). Among these methods some are specifically designed to explicitly account for space (e.g. Pavoine et al. 2011). However those studies generally consider spatially discontinuous community samples, i.e. the β -diversity component of Whittaker (1960), so that the underlying spatial framework can basically be related to the graph theory (Dale et al. 2002). When mapped data are concerned, such as for instance those on tropical trees provided by large permanent forest plots (Losos and Leigh 2004), the well-developed framework of the spatial point pattern analysis (Ripley 1976, 1977) offers an efficient means of studying the continuous spatial variation in species diversity (Condit et al. 2000; Shimatani 2001a; Rajala and Illian 2012). A synthesis of the

opportunities brought by this approach can be found in Wiegand et al. (2017).

Shen et al. (2013) and Pélissier and Goreaud (2015) concomitantly introduced within the framework of the spatial point pattern analysis a multi-type correlation function that provides a distance-dependent measure of species diversity accounting for non-uniform species differences. If both contributions independently arrived to very similar functions, we believe that to help ecologists to well-understand and to take full advantage of novel approach, further clarifications are needed, especially regarding how it is related to well-known functions of spatial point pattern analysis, namely Ripley's K-function (Ripley 1977) and its inter-type extension, the $K_{1,2}$ -function (Lotwick and Silverman 1982) on the one hand, and Rao's quadratic entropy on the other hand (Rao 1982).

The aim of this chapter is thus to introduce the rationale behind the functions of multivariate point pattern analysis that integrate between-species differences in a distance-dependent measure of diversity. It can basically be seen as an extension of the work of Shimatani (2001a) based on Simpson's entropy to Rao's entropy, one of the most powerful and flexible measure of species diversity (Pavoine 2012). It is also a particular implementation of the measurement of neighbourhood diversity which possesses some specific features due to the quadratic form of Rao's entropy. The methods section hereafter makes the synthesis between these originally independent developments and outlines the central role of Rao's entropy to reconcile them. We finally provide worked examples based on simulated data and on a real dataset from an experimental tropical forest plot in French Guiana.

2. Methods

2.1 Biodiversity measurement based on information theory

Definitions

Diversity accumulation curves

2.2 Distance-dependent entropy based on the point-process theory

Ripley's K

The framework of spatial point pattern analysis, as introduced by Ripley (1977), relies on the K-function, defined so that $\lambda K(r)$ is the expected number of neighbours located within a distance r of an arbitrary point of the pattern with intensity λ . Its classical unbiased estimator for a point

pattern located within a finite sampling window is

$$\hat{K}(r) = \frac{1}{\hat{\lambda}n} \sum_{i=1}^n \sum_{j=1}^n \mathbf{1}(\|x_i - x_j\| \leq r) c(i, j, r) \quad (1)$$

where n is the number of points of the pattern, $\hat{\lambda}$ is the pattern intensity generally estimated as n/A , with A the sampling window area. The indicator function $\mathbf{1}(\|x_i - x_j\| \leq r)$ equals 1 when points i and j are less than r apart, 0 else. $c(i, j, r)$ is correction of edge effects equal to 1 when the circle of radius r fits into the sampling window. When it does not, the isotropic correction proposed by Ripley (1977) applies to windows of any shape thanks to appropriate geometrical considerations (Goreaud and Pélissier 1999) to estimate the number of neighbors with no bias.

A common extension of this univariate function is the intertype (or bivariate) K_{12} -function proposed by Lotwick and Silverman (1982). Its estimator is:

$$\hat{K}_{1,2}(r) = \frac{1}{\hat{\lambda}_2 n_1} \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} \mathbf{1}(\|x_i - x_j\| \leq r) c(i, j, r) \quad (2)$$

$K(r)$ is a multiscale function of second-order neighbourhood properties of a point pattern, which allows robust tests of statistical significance against the null hypothesis of a random distribution of the points within the sampling window. In contrast, $K_{12}(r)$ allows characterizing the spatial interaction between the two types of points, which can be tested against various null hypotheses. The most interesting ones are the random labelling hypothesis, which means that, conditionally to the point locations, types 1 and 2 are randomly distributed among points of the pattern, and the population independence hypothesis, meaning that, conditionally to the species-specific spatial patterns, population of type 1 points is independent from population of type 2 points (Goreaud and Pélissier 2003).

A distance-dependent Simpson's entropy

Simpson's index of species diversity (Simpson 1949) gives the probability that a randomly selected pair of individuals in a sample is comprised of two different species. It is more than

an index: it is a measure of entropy (Jost 2006) and thus should be named Simpson's entropy. It is estimated with no bias by:

$$\hat{D} = 1 - \sum_{s=1}^S \frac{n_s(n_s - 1)}{n(n - 1)} \quad (3)$$

where n and S are the total numbers of individuals and species in the sample, respectively, and n_p the number of individuals of species s .

From this entropy, Shimatani (2001a) proposed a distance-dependent version based on Ripley's K-function, from which we can write a version standardized by $\hat{D}$ as

$$\hat{K}_S(r) = \left[1 - \sum_{s=1}^S \frac{\hat{\lambda}_s^2 \hat{K}_s(r)}{\hat{\lambda}^2 \hat{K}(r)} \right] / \hat{D} \quad (4)$$

where $\hat{K}(r)$ and $\hat{K}_s(r)$ are Ripley's K-functions for all points of the pattern (no distinction of species) and for points of species s , respectively, with $\hat{\lambda}$ and $\hat{\lambda}_s$ the corresponding intensity estimators.

The numerator of the above equation corresponds to function $\hat{\alpha}(r)$ of Shimatani (2001a), that we may name Simpson's neighbourhood entropy. Neighbourhood entropy (or diversity) is the entropy (or diversity) considered within a circle of radius r around each point. The interest of considering local diversity rather than at the scale of the whole window has emerged in the literature since Batty (1974). A review can be found in Altieri et al. (2021), with diverse approaches. It is not discussed that the functioning of a community is influenced by its biodiversity, but it is straightforward to consider that each individual is concerned by the diversity of its neighbourhood rather than of the whole community (e.g., Hubbell et al. 2001 showed that tree mortality increased with neighbourhood species richness in a tropical evergreen forest).

$\alpha(r)$ tends to D when r increases so $K(r)$ tends to 1. It is noticeable that $\hat{K}_S(r)$ has an expectation of 1 for all r under the random labelling hypothesis, i.e. a random distribution of the species names conditionally to the overall spatial pattern (Eckel et al. 2008).

Using the unbiased estimator of $\hat{K}(r)$ introduced in Equation 1, allows correcting for edge effects for sampling windows of any shape.

We can notice for the following that $\hat{K}_S(r)$ can also be estimated by considering the pairs of heterospecifics (Shimatani 2001a, eq. 9):

$$\hat{K}_S(r) = \left[\sum_{s=1}^S \sum_{t=1, s \neq t}^S \frac{\hat{\lambda}_s \hat{\lambda}_t \hat{K}_{s,t}(r)}{\hat{\lambda}^2 \hat{K}(r)} \right] / \hat{D} \quad (5)$$

with $\hat{K}_{s,t}(r)$ a generalization of $K_{12}(r)$ estimator introduced earlier, to any couple of different species s and t .

A generalization to Rao's entropy

Simpson's entropy is a special case of Rao's entropy, that is the expected divergence between individuals of a randomly selected pair in a sample (Rao 1982). An unbiased estimator (Shimatani 2001b) is:

$$\hat{H}_D = \sum_{s=1}^S \frac{n_s n_t}{n(n-1)} d_{s,t}$$

where $d_{s,t}$ is a measure of the pairwise divergence between species s and t (with $d_{s,t} = d_{t,s}$ and $d_{s,s} = 0$, but the triangular inequality $d_{s,t} + d_{s,u} \geq d_{s,u}$ that defines a distance is not required). H_D is a measure of diversity commonly used in ecology to take into account non-uniform distances between species, such as phylogenetic or functional distances (Pavoine and Bonsall 2011). Taking $d_{s,t} = 1$ for all $s \neq t$ and 0 otherwise, H_D reduces to Simpson's D above. We therefore straightforwardly propose to generalize $K_S(r)$ introduced above to a normalized, distance-dependent Rao's quadratic entropy $K_S(r)$ that integrates Ripley's correction of edge effects estimated by:

$$\hat{K}_R(r) = \left[\sum_{s=1}^S \sum_{t=1, s \neq t}^S \frac{\hat{\lambda}_s \hat{\lambda}_t \hat{K}_{s,t}(r) d_{s,t}}{\hat{\lambda}^2 \hat{K}(r)} \right] / \hat{H}_D$$

This function has an expectation of 1 under the random labelling hypothesis and gives the mean proportion of Rao's sample diversity (i.e., H_D) in a disc of radius r centred on an arbitrary point of the pattern. $\hat{K}_R(r)$ reduces to $\hat{K}_S(r)$ when $d_{s,t} = 1$ for all species $s \neq t$.

Multitype correlation and β diversity

Our K_R - and K_S -functions are cumulative functions as is the original K-function of Ripley (1977).

This means that they are estimated for a series of concentric neighbourhood discs of increasing radii, for instance by a regular distance interval dr . Some authors however prefer to refer to the framework of the pair-correlation function, $g(r)$ (Stoyan et al. 1987; Wiegand and Moloney 2004), that can be seen as the derivative of $K(r)$, giving the probability to encounter neighbouring points in concentric annuli between r and $r + dr$. $K_d(r)$ of Shen et al. (2013) is such a function, very close to $g_R(r)$, the derivative of our $K_R(r)$ above (Pélissier and Goreaud 2015). $K_d(r)$ is therefore an awkward appellation for Shen's et al. function, which would have been better named as $g_d(r)$ as to remain consistent with the classical notation used in point pattern analysis.

The same reasoning can be built from the so-called Simpson's β diversity index to $g_R(r)$. Chave and Leigh (2002) defined the $F(r)$ function as the probability for two individuals located r apart to belong to the same species. Shen et al. (2013) named $1 - F(r)$ the 'spatially explicit Simpson's index of diversity', denoted $\beta(r)$ by Shimatani (2001a), that is the probability for two individuals located r apart to belong different species. We will rather write $\beta_S(r)$ for consistency with our notation system. It can be straightforwardly be generalized to $\beta_R(r)$, the average divergence between to individuals taken at distance r , thus a spatially explicit Rao's index of diversity. These measures are actually β differentiation diversity measures sensu Jurasinski et al. (2009): they characterize how different spatial units r apart are.

We can thus define $g_S(r)$, a normalized version of $\beta_S(r)$ and $g_R(r)$, its generalization to Rao's entropy.

2.3 Synthesis: a consistent measure of neighbourhood diversity

2.4 Testing ecological models

The K_R -function allows computing phylogenetic or functional entropy in the neighbourhood of individuals with respect to distance. Beyond measuring diversity, it also allows inferring ecological processes responsible of the observed patterns by testing them against appropriate null hypotheses.

The random labelling null hypothesis

The null hypothesis of random labelling originally proposed by Shimatani (2001a) provides a test of departure of $\hat{K}_R(r)$ from a random distribution of the mark labels among points of the

pattern. It allows testing for an absence of interaction between neighbouring points, conditionally to the overall pattern of point locations, so that the probability to find a pair of identical marks is independent of distance. When the function is above 1, the mean diversity within a disc of radius r from an arbitrary point of the pattern is higher than the total sample diversity, and conversely. Local confidence limits of the random labelling hypothesis at a given significance level can be built from Monte Carlo rank tests Diggle (1979). But this hypothesis basically relies on the assumption that the marks result from a random process that affects individuals a posteriori with regards to the process of point location, so that we have for all s types of marks $K_s(r) = K(r)$ (Goreaud and Pélissier 2003; Eckel et al. 2008) and hence $K_R(r) = 1$. This could for instance corresponds to a multitype process resulting from a disease attack in a plant population that would mark the individuals as sick, non-infected or healthy carrier (see also Eckel et al. 2008 for an example in socio-economics). Sounded reasons for testing the random labelling hypothesis are however hardly conceivable in the context of species diversity, so that testing this hypothesis only permits limited inferences about the processes at the origin of the observed pattern of species diversity.

The population independence null hypothesis

A more meaningful hypothesis for species diversity studies is the one of mutually independent populations, postulating that the spatial pattern of each type of mark (or population) is the realization of a specific spatial process (for instance of limited dispersion) and that the location of points of each population is independent from the location of the points of the other populations in the resulting multivariate pattern. This is a generalization of the population independence hypothesis proposed for testing the bivariate $K_{1,2}$ -function by Lotwick and Silverman (1982) for which we expect that $K_{s,t}(r) = \pi r^2$ for any marks $s \neq t$ (Eckel et al. 2008), and hence, here also, $K_R(r) = 1$. Note that this hypothesis differs from the random labelling hypothesis, except when all specific spatial patterns result from a stationary Poisson point process (Diggle 1983).

A classical method for testing the mutually independent population hypothesis is to build a Monte Carlo rank test keeping the pattern of each population unchanged, but randomizing their relative

positions by a random shifting procedure of each pattern around a torus (Goreaud and Pélissier 2003). Such a procedure is cumbersome for highly multivariate patterns and is thus abandoned.

The species equivalence null hypothesis

More interestingly $K_R(r)$ provides opportunities for testing a highly relevant ecological hypothesis for diversity studies, based on $d_{s,t}$, which quantifies how species differ one from each other. Shuffling between-species distances a large number of times means eliminating their differences and therefore testing a null hypothesis of species equivalence, put forward, for instance, in the neutral theory of biodiversity (Hubbell 2001). This null hypothesis can be tested conditionally to the pattern of point locations and to the spatial distribution of the species labels by permuting simultaneously the rows and columns in $d_{s,t}$, which is equivalent, in the case of a cophenetic distance matrix, to permuting the tips of a hierarchical (taxonomic or phylogenetic) tree (Hardy 2008). It is noteworthy that under the null hypothesis of species equivalence, $K_S(r)$ provides a theoretical expectation of $K_R(r)$, i.e. the value expected when $d_{s,t}$ is a uniform between-species distance matrix. When focusing on the species equivalence hypothesis, $K_R(r)/K_S(r)$, the standardization proposed by Shen et al. (2013), provides a constant expectation of 1 at all distances. With reference to these authors and to the usual notation in point pattern analysis, this ratio is appropriately designated as $K_d(r)$.

3. Application

3.1 Simulated data

We provide here a few examples to illustrate how the K_R -function behaves.

Habitat preference

In a first experiment (Figure 1), we simulated habitat preferences by generating heterogeneous spatial patterns in a checked 1-ha plot with two habitats: eight species were simulated with a twofold higher intensity in habitat 1, and 8 species with a twofold higher intensity in habitat 2, so that species abundant in one habitat were rare in the other one. We independently simulated the species-specific spatial patterns as heterogeneous Poisson or heterogeneous Neyman-Scott processes with a Cauchy dispersal kernel (Ghorbani 2013) mimicking limited dispersion. In the latter we fixed the scale parameter of the Cauchy kernel

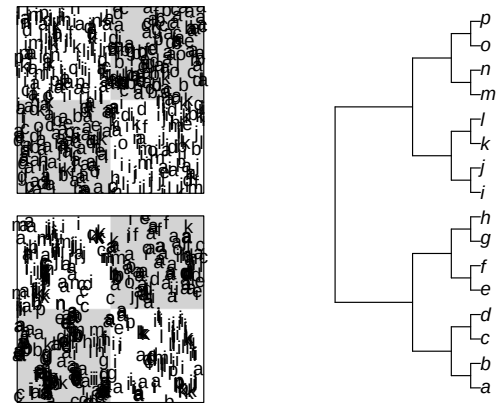


Figure 1. Virtual heterogeneous spatial patterns (left) mimicking habitat preferences of 16 species in a checked 1-ha plot with two habitats: species *a* to *h* have a twofold higher habitat preference in grey patches and species *i* to *p* a twofold higher habitat preference in white patches. The random pattern (top) is obtained by the superimposition of heterogeneous species-specific Poisson processes. The pattern of limited dispersion (bottom) is obtained by the superimposition of species-specific Neyman-Scott processes with a Cauchy kernel (see text for details). The hypothetical phylogenetic tree (right) simulates trait clustering: species close in the tree have identical habitat preferences.

at 1, meaning that only 5% of offsprings were dispersed beyond around 20 m from their parents (Munoz et al. 2013), so that most of them were located within the sampling window. The 16 species-specific patterns were then assembled into multivariate spatial patterns. A hypothetical phylogeny mimicking trait clustering was simulated, where species close in the tree have same habitat preferences (Hardy and Senterre 2007).

Figure 2 shows that both these patterns exhibit a significant negative departure from the null hypothesis of species equivalence for the same range of r values that recovers the scale of the phylogenetic spatial structure representing the simulated pattern of habitat preferences.

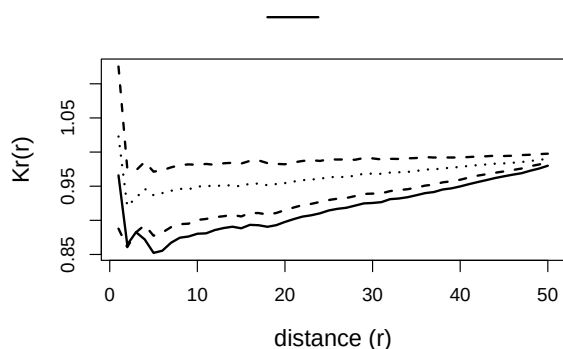


Figure 2. Test of the species equivalence hypothesis. $K_R(r)/K_S(r)$ curves are computed with the cophenetic distance matrix derived from the clustered phylogenetic tree and tested against a 1% confidence envelop of the species equivalence hypothesis obtained from 1000 Monte Carlo simulations.

Limited similarity

In a second experiment, we simulated a pattern of competitive interaction within a fixed range of 5 m around each individual, using a multi-Strauss process with, as between-species interaction matrix, the cophenetic distance derived from the clustered tree above. This means that the between-species competition intensity within a neighbouring disc of 5 m is proportional to their distance within the phylogenetic tree: species that are close together in the tree compete less than species that are far in the tree.

Figure 4 shows that the resulting pattern of the point locations, as revealed by $\hat{K}(r)$, is significantly regular with a deep negative peak at $r = 5$ m and that $K_R(r)/K_S(r)$ departs significantly and

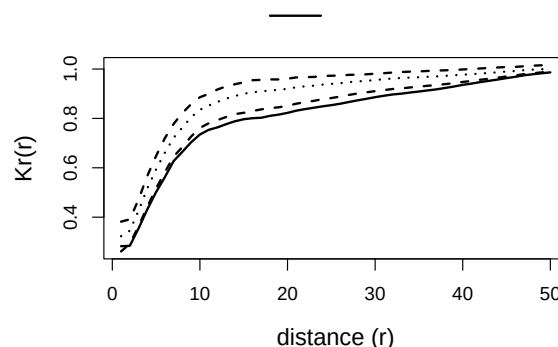


Figure 3. Test of the species equivalence hypothesis. $K_R(r)/K_S(r)$ curves are computed with the cophenetic distance matrix derived from the clustered phylogenetic tree and tested against a 1% confidence envelop of the species equivalence hypothesis obtained from 1000 Monte Carlo simulations.

positively from the null hypothesis of species equivalence for the same range of distances, confirming the limiting similarity pattern originally simulated. Restricted permutations within a cophenetic distance matrix, i.e. between species within clusters or clades in a hierarchical tree, can provide more specific tests of evolutionary hypotheses (Hardy 2008).

All the functions introduced above along with resources for computing tests of statistical significance based on Monte Carlo simulations have been implemented within the ads package for R (Pélissier and Goreaud 2015). Samples of R code used in the Application section are given in Appendix.



Figure 4. Repulsive point pattern

3.2 Paracou: a tropical forest plot in French Guiana

Study site and data

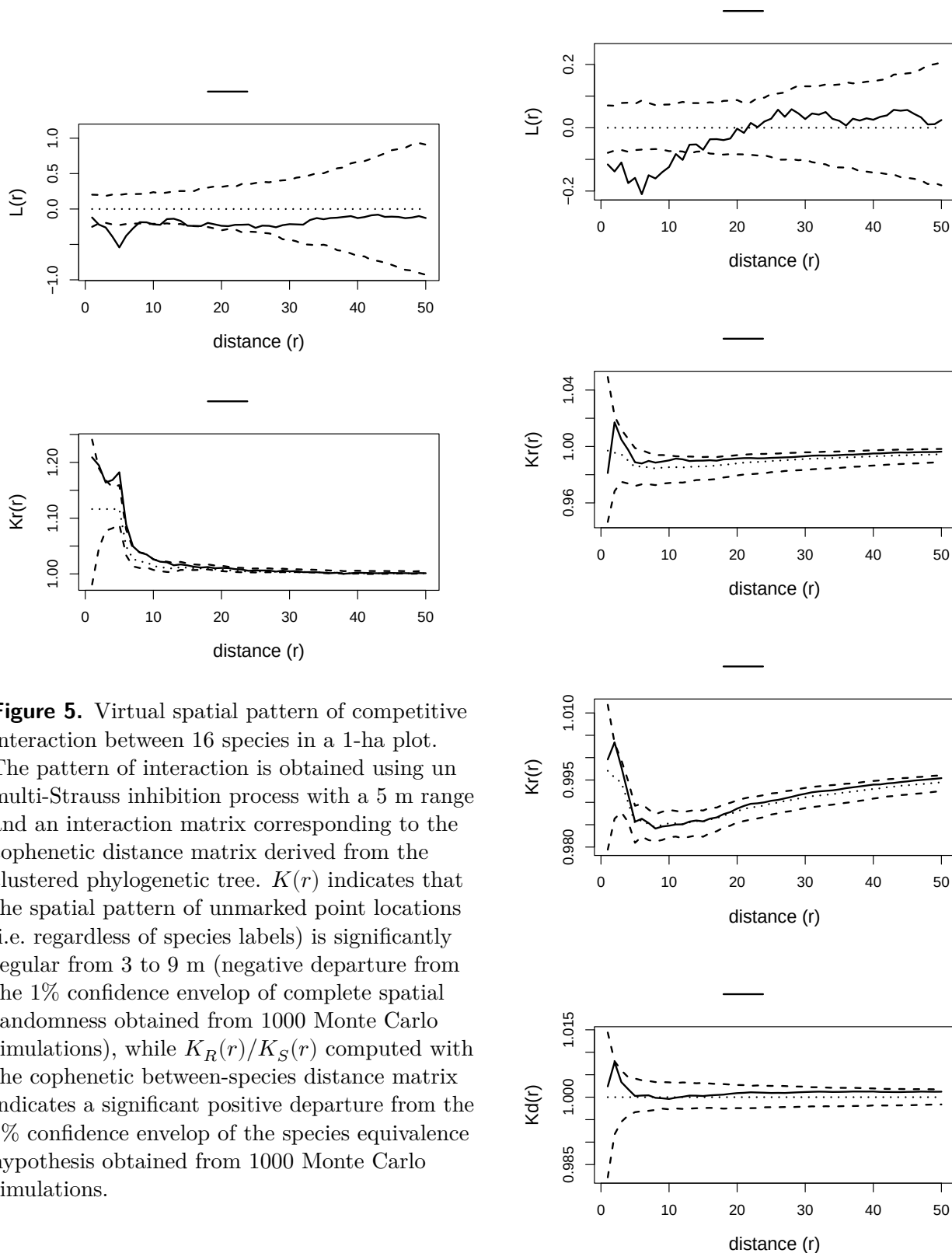


Figure 5. Virtual spatial pattern of competitive interaction between 16 species in a 1-ha plot. The pattern of interaction is obtained using an unmarked multi-Strauss inhibition process with a 5 m range and an interaction matrix corresponding to the cophenetic distance matrix derived from the clustered phylogenetic tree. $K(r)$ indicates that the spatial pattern of unmarked point locations (i.e. regardless of species labels) is significantly regular from 3 to 9 m (negative departure from the 1% confidence envelop of complete spatial randomness obtained from 1000 Monte Carlo simulations), while $K_R(r)/K_S(r)$ computed with the cophenetic between-species distance matrix indicates a significant positive departure from the 1% confidence envelop of the species equivalence hypothesis obtained from 1000 Monte Carlo simulations.

We used for application a dataset of tropical forest trees in Paracou experimental station, French Guiana (Gourlet-Fleury et al. 2004). Paracou

was set up in 1984 to study the impact of silvicultural treatments on the forest dynamics based on control and logged-over plots. We used here data from a control plot of 250 x 250 m settled in 1990, in which `n_trees` trees with minimum stem diameter at breast height (dbh) of 10 cm were mapped and identified to species level (335 different species; Fig. 3-topleft). Since After several scientific programs developed in French Guiana, a database of functional traits is available for the sampled species. For the present application we selected the basic functional traits defining the Leaf-Height-Seed (LHS) scheme of plant strategies (Westoby 1998), i.e. the specific leaf area (SLA in $\text{cm}^2.\text{g}^{-1}$ of dry matter), the expected tree height at maturity (H in m) and, as a proxy for seed mass, a seed size ordered-codification (Seed), from 1 (very small seeds) to 5 (very big seeds). These traits are acknowledged to illustrate the major trade-offs in plant fitness between resource capture and allocation: SLA characterizes the importance of the growth response to favourable conditions, while H and Seed characterizes the importance of investment in the capacity to respond to disturbance. We also included the wood specific gravity (WSG in g.cm^{-3} of dry matter) as a fourth trait specifically accounting for additional important functions in trees, such as water transport, structural support, resource storage, defence, etc. (Chave et al. 2009). We built from these data, two different between-species distance matrices, a phylogenetic one based on the APG III classification (Chase and Reveal 2009) and a functional one based on the four traits described above. The phylogenetic distance was simply computed from APG III hierarchical tree, considering $d_{p,q} = 1$ if species p and q were from the same genus, $d_{p,q} = 2$ if they were from different genera in the same family and $d_{p,q} = 3$ if they were from different families. Such a cophenetic distance matrix has ultrametric properties ensuring desirable maximization properties to Rao quadratic entropy (Pavoine et al. 2005). To ensure the same ultrametric properties to the functional distance matrix, we computed the cophenetic distance for a hierarchical clustering with Ward method (i.e. minimizing total within-cluster variance) based on functional trait dissimilarities, taken as Gower distance, as recommended for mixed variables (Gower 1971). Both distance matrices were finally rescaled by their maximum in order to keep HD within the 0-1 standard range of values for diversity indices.

Analysis of species diversity spatial pattern

The L-function (Fig. 3-topright), i.e. the linearized univariate version of Ripley K-function, shows a significant departure towards regularity of the unmarked spatial pattern (i.e. regardless of species labels) up to 9 m (negative departure from the 1% confidence envelop of complete spatial randomness obtained from 1000 Monte Carlo simulations). $K_R(r)/K_S(r)$ computed with a phylogenetic between-species distance matrix shows a significant positive departure from the null hypothesis of species equivalence at short distances (based on 1000 Monte Carlo simulations), indicating limiting similarity (Fig. 3-bottomleft). As a short-range spatial regularity in tree stands is generally interpreted as resulting from self-thinning due to intense local competition (e.g. Kenkel 1988), it therefore seems that the sharing of resources between locally competing neighbours enhanced niche differentiation and thus species diversity. This pattern doesn't exist however when considering the between-species distance matrix based on the 4 functional traits together (Fig. 3-bottomright) or on each of them separately (Fig. S2), though these traits are generally considered as phylogenetically conserved (e.g. Baraloto et al. 2012). This probably means that niche trade-offs that caused short-range limiting similarity do not concern the four functional traits considered in the present study.

4. Conclusion

We demonstrated in this paper that the function of multitype point pattern analysis concomitantly proposed by Shen et al. (2013) and Pélissier and Goreaud (2015), which integrates between-species differences in a distance-dependent measure of diversity, relies on two well-known analytical frameworks: the spatial point pattern analysis popularized by Ripley (1977) through his K-function, and the quadratic entropy introduced by Rao (1982) as a diversity coefficient accounting for non-uniform between-species distances. It follows that a spatially continuous and unbiased measure of species diversity can be envisioned as a simple wrapper of classical functions of point pattern analysis, namely the K- and K12-functions. Introducing a non-uniform between-species distance matrix in the quantification of species diversity also provides opportunities to consider different and potentially complementary metrics, based on either taxonomic, phylogenetic or functional differ-

ences between species. Observed spatial diversity patterns can thus be tested against the highly meaningful ecological hypothesis of species equivalence emphasized by the neutral theory of biodiversity (Hubbell 2001). This new tool provides a powerful and robust tool to detect significant trait clustering or divergence expected to arise in ecological communities under the driving forces of habitat preferences or local competitive interactions.

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